

Amit Kumar Pandey

+91 9155840605 | pandeyamit2871@gmail.com |

[LinkedIn](#) | [Github](#) |

Languages: Python, JavaScript, TypeScript, SQL

Data Science & ML: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost, CatBoost Regression, Classification, Clustering, Model Evaluation, EDA, Feature Engineering, Model Serialization (pickle/dill)

Web Development: Flask, React, Express, Node.js, REST APIs, MongoDB, PostgreSQL, HTML, CSS, Unicorn/WSGI

Cloud & DevOps: AWS (EC2, Elastic Beanstalk, CodePipeline, IAM), CI/CD Pipeline Design, Git, GitHub

Tools: Jupyter Notebook, Google Colab, Supabase, Netlify, Vercel, Render

Mathematics & Statistics: Probability, Statistics, Linear Algebra

INTERNSHIP

Web Developer | Seat Mentor

June 2026 – Present

- Built the company website from the ground up—architected and developed both frontend and backend, taking it from concept to a fully functional, production-deployed product.
- Owned end-to-end backend development and ongoing management, including API design, database integration, and server maintenance, ensuring 99%+ uptime.
- Independently translated business requirements into technical specifications and shipped features with minimal oversight over a 1+ year engagement.

Data Analytics | GlobCred

May 2025 – July 2025

- Cleaned and pre-processed data using Pandas and visualized it using Matplotlib/Seaborn.
- Performed Exploratory Data Analysis (EDA) on a sales dataset to identify insights and patterns.
- Built an interactive sales analytics dashboard to track key performance metrics and trends.
- Tech stack: Tableau (visualization), Pandas (data cleaning), Matplotlib, Seaborn (visualization).

PROJECTS

Student Performance Predictor — End-to-End ML Web App with Automated AWS CI/CD

[Live Project](#)

- Built and deployed a Flask web application that predicts student exam performance from demographic and study-habit inputs, using a regression pipeline built with Scikit-learn, XGBoost, and CatBoost.
- Designed and shipped a fully automated CI/CD pipeline (GitHub → AWS CodePipeline → Elastic Beanstalk) so every push to main triggers an automatic build-free deploy straight to a live Python 3.13 / Amazon Linux 2023 environment.
- Diagnosed and fixed a platform-level Elastic Beanstalk configuration bug (an obsolete .ebextensions namespace) and an IAM permissions gap (missing elasticbeanstalk:CreateApplicationVersion) that were silently breaking deployments.
- Configured least-privilege IAM roles across three services — CodePipeline, Elastic Beanstalk service role, and the EC2 instance profile — to support secure, repeatable deployments.
- Tech stack: Python 3.13, Flask, Unicorn, Scikit-learn, XGBoost, CatBoost, Pandas, NumPy, AWS (Elastic Beanstalk, CodePipeline, IAM), GitHub.

Video Compressor — Privacy-First, Zero-Upload Video Compression Tool

[Live Project](#)

- Built a full-stack web app that compresses videos entirely client-side using FFmpeg compiled to WebAssembly, eliminating file uploads entirely—saving users gigabytes of bandwidth and ensuring zero exposure of video content to any server.
- Implemented target-file-size compression by calculating required video bitrate from file duration, letting users compress to a specific output size instead of a generic quality preset.
- Engineered cross-origin isolation (COOP/COEP headers) to enable SharedArrayBuffer, unlocking multi-threaded WASM performance with real-time progress tracking during compression.
- Set up a full CI/CD pipeline using AWS CodePipeline, CodeBuild, and Elastic Beanstalk, automating build and deployment on every push.
- Tech stack: React, Vite, TailwindCSS, Express, FFmpeg (WASM), TypeScript, AWS (Elastic Beanstalk, CodePipeline, CodeBuild).

ACADEMIC RECORD

- BCA – Data Science & Machine Learning, Sri Balaji University, Pune — 9.00 CGPA till sem 4 (2024–present)
- HSC – Science, Jharkhand Academic Council — 77% (2021–23)
- SSC – Regular, Central Board of Secondary Education — 87% (2019–21)

EXTRA CURRICULAR ACTIVITIES

- Secured 1st position in multi-stage hackathon competition – CodeFusion 2k26
- Secured 1st place in an Intra-college Essay Writing Competition
- Finalist for Google Build with AI – Agentic Premier League
- Participated in Technovate competition – 3rd Rank in Technical MCQ Round
- Collaborated with a class-wide team to secure 3rd position among 16 competing classes in Fresher 2k24